Fly Brain Lab

Interactive exploration of the complete adult Drosophila melanogaster central nervous system on the BANC v888 connectome

Every neuron is real. 188,508 cells at their true soma positions, 2,819,440 cell-type pairs of real connectivity, real neuropil boundaries, real morphology, real measured locomotion - with a leaky integrate-and-fire simulation layered on the real wiring.

Bates, Phelps, Kim, Yang, et al. (2026). Distributed control circuits across a brain-and-cord connectome. Nature. DOI: 10.1038/s41586-026-10735-w

Data snapshot: Harvard Dataverse doi:10.7910/DVN/7WTH1N - portal: banc.community - viewer: ng.banc.community/view

This document and the software are a research/education tool. Sections marked MODEL describe approximations - see 'What is real vs modeled' before citing dynamics as biology.

1. Abstract & Purpose

Fly Brain Lab is a locally runnable / statically hostable scientific visualization of the BANC (Brain And Nerve Cord) connectome - the first synapse-resolution reconstruction of a fruit fly's complete central nervous system: brain, suboesophageal/gnathal region, cervical connective and ventral nerve cord (VNC). The application shows the full ~188,508-cell dataset as a navigable 3D point cloud with real neuropil boundaries, a full-screen 'live brain' view that streams the active fly's neural activity between browser tabs, on-demand real neuron skeletons (SWC) fetched from the BANC public repository, and a simplified 3D world where a simulated fly's movement is decoded from real motor neuron pools - or replayed from the actual BANC specimen's recorded locomotion.

2. Data Provenance

Item	Value
Dataset	BANC - Brain And Nerve Cord connectome
Version	v888 materialization (snapshot 2026-04-17), the paper snapshot
CNS cells	188,508 rows in banc_888_meta.feather
Neuron-level connections	13,620,865 (edgelist_simple_v3)
Cell-type pairs loaded	2,819,440 - ALL pairs, lossless
Cell types	11,497
Predicted synapses	~199M (v3 table, >=10-voxel threshold)
NT classes	8-class predictions + unknown
Species / stage	D. melanogaster adult female, whole CNS

Version notes (for citation)

* v626 = preprint snapshot - superseded by v888.
* v888 = reproducible, DOI-citable release used here.
* FlyWire Codex hosts a live/evolving version - do not confuse it with this fixed snapshot.
* Paper analysis used synapse table v2 (>=5 voxels). This project uses v3 (>=10 voxels),
* the project's recommended default for new analyses. Both remain in the source bucket.

Links

* Paper (Nature): doi.org/10.1038/s41586-026-10735-w
* Data snapshot (Harvard Dataverse): doi.org/10.7910/DVN/7WTH1N
* Project portal: banc.community
* Neuroglancer viewer: ng.banc.community/view
* Live codex: codex.flywire.ai
* Source bucket: gs://lee-lab_brain-and-nerve-cord-fly-connectome

3. Architecture

Static web app: Vite + TypeScript + Three.js. Python pipeline exports the public BANC data into compact browser-served artifacts. No backend.

```
index.html      main app - 3D world (fly + flag) + CNS panel
brain.html      full-screen CNS viewer tab (live-synced via BroadcastChannel)
src/main.ts     app init, UI, live-activity broadcaster
src/brain_view.ts  standalone brain tab - sync, skeletons, metrics, info
src/banc_loader.ts BANC JSON/binary loading, NT + region maps
src/connectome.ts  cell-type connectome construction
src/brain_data.ts  188K neuron cloud + region geometry
src/neural_viz.ts  Three.js instanced rendering, filters, SWC
src/neural.ts      LIF network + dopamine-gated plasticity
src/sensory.ts     visual-field encoding -> real lamina inputs
src/motor.ts       real motor/DN activity -> movement decode
src/experiment.ts  experiment loop + real trajectory replay
src/world.ts       3D world, fly model, flag, movement trail
src/reward.ts      reward/dopamine system
src/config.ts      constants
public/data/   exported research data (neurons/types/pairs/regions/metrics/trajectory)
scripts/       download_banc_data.py, export_realism_data.py, make_pdf.py
```

Binary connectivity format (banc_connections.bin, 'BCN1')

```
magic   : u8[4] = 'B','C','N','1'
nTypes  : u32    | nPairs : u32
types   : u16[nTypes] name lengths + utf8 names
pairs   : nPairs x { pre u32 | post u32 | total u32 |
                     connCount u32 | mean f32 | median f32 }
```

2,819,440 pairs = 64.6 MB - streamed and decoded with typed arrays; the earlier 539 MB JSON of the same data was never usable in a browser.

3b. Scientific Methodology

Source data & materialization

Pipeline consumes the BANC v888 materialization (snapshot 2026-04-17) - the fixed, DOI-citable release of the Nature publication - from the consortium's public GCS bucket:

```
banc_888_meta.feather              188,508 cells x 81 annotation cols
banc_888_edgelist_simple_v3        13,620,865 connections (>=10 voxels)
banc_888_metrics.feather           per-cell morphometrics
banc_banc_space_split_swc/*.swc    proofread skeletons (lazy fetch)
behavior/BANC_fly_behavior/        real BANCr Y-maze locomotion
```

Network reduction

One representative node per annotated cell type (11,497 types) at the true soma centroid; hemisphere & majority neuropil from member cells. Glia/trachea/non-neuron rows (~13K) excluded from neurons and network. All meta rows and all cell-type pairs are preserved losslessly in the exported artifacts.

Validation performed

* Hemisphere sign verified vs 'side' column (left = +x) - fixed an inversion
* Region envelopes cross-checked vs per-root_region counts
* Full counts preserved end-to-end (188,508 rows; 2,819,440 pairs)
* 19,193 missing-position rows excluded, not fabricated
* Binary container round-trips all four pair statistics

4. The Mathematics

4.1 Coordinate normalization

```
center_i = (min_i + max_i)/2   (per axis, zero sentinels ignored)
scale    = max(range_x, range_y, range_z)/2   = 537,312 nm

display.x = (x - x_center)/scale    lateral      (+x = anatomical LEFT)
display.y = -(y - y_center)/scale   longitudinal (brain up, VNC down)
display.z = (z - z_center)/scale    dorsoventral (dorsal = +z)
```

Verified against the real 'side' column: side=left cells sit at +x and side=right at -x (BANC axis0 is lateral and positive = left). The same norm constants are applied to fetched SWC skeletons so morphologies overlay the point cloud exactly.

4.2 Neuropil geometry

```
per region x side:  center = mean(points);  semi-axes = 2*std(points)
VNC + cervical connective rendered as real cylinders along the longitudinal axis.
```

4.3 LIF dynamics

```
dV/dt = (V_rest - V)/tau_decay
      + Sum_spikes w_syn * s(nt) * delta(t - t_arr)
      + I_noise(class, NT) + I_injected

if V >= V_thresh:  spike; V -> V_reset; refractory t + tau_ref
```

4.4 Signed synaptic effect by predicted NT

NT	Effect (fly biology)	sign x scale

acetylcholine	excitatory (nAChR)	+1.0
GABA	inhibitory (Rdl/GABA-A)	-1.0
glutamate	inhibitory (GluCl, dominant fast inhibitory channel)	-0.9
histamine	inhibitory (HisCl, photoreceptor output)	-1.0
dopamine	modulatory	+0.15
serotonin	modulatory	+0.10
octopamine	modulatory	+0.15
tyramine	modulatory	+0.15
unknown	most CNS cells are cholinergic	+0.40

4.5 Conduction delay

```
delay = 0.5 ms (synaptic latency) + dist(pre,post) x 0.72 ms
(1 unit ~ 537 um at ~0.75 m/s insect conduction)
```

4.6 Dopamine-gated plasticity (KC -> MBON)

```
Delta w = eta * DA(t) * a_KC(t) * dt      (clamped [0.01, 5])
```

Plasticity is restricted to Kenyon-cell -> MBON synapses - the canonical mushroom-body associative-learning motif - driven by a global decaying dopamine level released by reward events.

4.7 Sensory -> motor decode

```
forward = clamp((avgLeg*0.7 + avgWing*0.3)*5, 0, 1)
turn    = clamp((avgLegR - avgLegL)*8, -1, 1)    (real hemispheres)
reach   = clamp(avgDN*6, 0, 1)
```

Inputs target real lamina output neurons (L1-L5, Mi1, Tm*, C2/C3, Lawf*) and real R7/R8 photoreceptors; motor readout pools real leg_motor_neuron / wing_motor_neuron / descending_neuron cell classes.

5. What You Can Do

* 188K-neuron instanced cloud - real positions, class colors, live activity brightness.
* Real neuropil wireframes - per-region bounding ellipsoids/tubes, left/right correct.
* Full-screen brain tab - the ACTIVE fly's live brain streamed over BroadcastChannel.
* Deep zoom 0.03-30x; region explanations; real neuron counts per neuropil.
* Click a neuron -> real BANC annotation + real morphology metrics + real SWC skeleton.
* Region filters (optic lobe, central brain, mushroom body, antennal lobe, SEZ, neck, VNC).
* Fly movement trail in the world; real BANCr Y-maze trajectory replay (toggle button).
* Region pulse bars, learning/dopamine/activity graphs.

6. Real vs. Modeled (honest table)

Component	Status
Neuron positions	REAL - BANC soma coordinates (unlocalized excluded)
Counts / types / classes	REAL
Neuropil regions & sides	REAL root_region / side annotations
Wireframes	REAL-DERIVED 2-sigma bounds (not EM meshes)
Connectivity	REAL v3 edgelist, lossless, all pairs
Synapse signs	REAL-INFORMED via predicted NTs
Skeletons	REAL per-cell SWC (subset coverage)
Morphology metrics	REAL (banc_888_metrics)
Fly locomotion (replay)	REAL measured BANCr track
Spiking dynamics	MODEL - LIF + injected currents
Sensory/motor gains	MODEL (real pools, simplified encode)
World / flag / fly body	INVENTED apparatus (labeled in code)

7. Under Research & Development

* One sim neuron per cell type - inter-type realism kept, intra-type variance collapsed;
* synaptic strengths are pair-means, not per-connection counts.
* No dendrite-vs-axon synaptic weighting; morphology not used in coupling.
* Only cells with a released split-SWC render a skeleton.
* Wireframes bound the soma cloud - true EM neuropil surfaces are future work.
* NTs are predictions; a fraction differ from ground-truth staining.
* Delays use a single conduction velocity constant.
* The behavior loop is a sandbox - reasonable defaults, not fitted physiology.

8. Reproduce / Regenerate

```
npm install            # three, vite, typescript
npm run dev            # localhost:3000  (brain tab /brain.html)

pip install pandas pyarrow numpy
python scripts/download_banc_data.py     # ~415 MB cached to data/raw/
python scripts/export_realism_data.py    # norm + metrics + trajectory
python scripts/make_pdf.py               # this document

Deploy: vercel (Vite preset auto-detected; output dist/).
public/data (~135 MB) is copied into dist and served statically.
```

8b. Glossary

Term	Meaning
Connectome	Complete map of neurons and their synaptic connections
Materialization	Fixed queryable snapshot of EM segmentation + annotations
Neuropil	Named synaptic region (medulla, MB calyx...) - our wireframe unit
Neuromere	Segmental VNC unit (T1-T3 thoracic, A1-A10 abdominal)
Soma	Cell body - the dots rendered are real soma positions
root_region	Ito-atlas neuropil code annotating each cell's dendritic home
super_class / cell_class / cell_type	Coarse->fine BANC annotation hierarchy
Hemilineage	Developmental origin class (paired with NT identity)
KC / MBON / DAN	Kenyon cell - MB output neuron - dopaminergic neuron
LIF	Leaky integrate-and-fire - the spiking model used
GluCl	Glutamate-gated Cl channel - why fly glutamate inhibits
NBLAST	Morphological similarity metric used for cell typing
DN / AN	Descending / ascending neuron - brain<->cord tracts

9. References & Links

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